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Retinal Imaging Biomarkers and Correlation to Systemic Disease Activity in Pediatric Sickle Cell Disease

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Purpose: To correlate retinal imaging findings with systemic disease activity in children with sickle cell disease (SCD).

Design: A retrospective consecutive series.

Subjects: Children with SCD aged ≤18 years who had an ophthalmic examination at Boston Children's Hospital between January 1998 and August 2022 were included.

Main Outcome Measures: Systemic findings included the number of hospitalizations, number of strokes, treatment with hydroxyurea, hemoglobin (Hgb), and fetal Hgb levels, and time-averaged mean velocity (TAMV) in the right middle cerebral artery (RMCA) and left middle cerebral artery (LMCA) on transcranial Doppler (TCD).

Methods: Total retinal thickness was measured on macular OCT (Spectralis OCT2, Heidelberg Engineering). Vessel density (VD) of the superficial capillary plexus (SCP) and deep capillary plexus (DCP) and superficial foveal avascular zone area were measured on 6 × 6-mm OCT angiography (OCTA) scans.

Results: Six hundred six eyes from 303 pediatric SCD patients (53% males) were included. OCT and OCTA images were acquired on 104 (17.2%) and 60 (9.9%) eyes at presentation and on 159 (26.2%) and 100 (16.5%) eyes at the final visit, respectively. When adjusting for race and age, retinal thinning on OCT was associated with a higher frequency of hospitalizations, a higher frequency of strokes, and treatment with hydroxyurea. Retinal thickness in the inferior and temporal macula was positively correlated with TAMV in RMCA and in LMCA. Foveal retinal thickness was positively correlated with Hgb level. Similarly, reduced VD in the SCP and DCP in the inferior temporal macula correlated with a higher number of hospitalizations and strokes. A higher VD of the DCP in the inferior-temporal macula positively correlated with TAMV in RMCA ($\rho = 0.328$, $P = 0.3$) and in LMCA ($\rho = 0.342$, $P = 0.029$). A higher Hgb level correlated with a higher prevalence ($\rho = 0.237$, $P = 0.037$) and severity ($\rho = 0.299$, $P = 0.008$) of peripheral retinopathy in HbSC, while it correlated with lower prevalence ($\rho = -0.183$, $P = 0.004$) and severity ($\rho = -0.185$, $P = 0.004$) of peripheral retinopathy in HbSS genotypes. Visual acuity did not correlate with TCD velocity, Hgb level, or number of hospitalizations in HbSS or HbSC genotypes.

Conclusions: OCT and OCTA findings are correlated with the severity of systemic disease in children with SCD. Imaging parameters were better correlated with key outcomes such as stroke and hospitalizations than visual acuity. The results suggest that quantitative measures on retinal imaging could be used as biomarkers to predict systemic disease risk and activity.

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Sickle cell disease (SCD) is an inherited blood disorder that affects millions globally, with a particularly notable impact on the pediatric population.[1,2] Children with SCD are susceptible to various complications, including sickle cell retinopathy and sickle cell maculopathy, both characterized by decreased retinal vascular flow and vascular occlusion, which may result in vision loss.[3,4] In addition, cerebral complications such as ischemic stroke have a great impact on the morbidity and survival of patients with SCD.[5] Transcranial Doppler (TCD) is a valuable tool in identifying patients at increased risk for ischemic brain injury. Notably, elevated time-averaged maximum mean velocity (TAMV) with values >200 cm/second in the middle cerebral artery was associated with an increased risk of stroke.[6,7] Lower fetal hemoglobin (HgbF) was associated with an increased risk of admission to the intensive care unit.[8]

The relationship between retinal and systemic findings is an emerging area of interest, particularly in understanding how changes in retinal vasculature can reflect underlying systemic conditions. A recent study showed that patients with silent cerebral infarctions on magnetic resonance imaging had lower vessel density (VD) in the deep capillary plexus (DCP) compared with those without infarctions,

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suggesting that changes in retinal blood flow and vessel tortuosity may serve as indicators of systemic disease severity, particularly in the brain.[9] However, to date, no studies have analyzed the correlation between retinal imaging and systemic findings, including blood flow velocity on TCD, number of hospitalizations, number of strokes, treatment with hydroxyurea, and hemoglobin (Hgb) level in pediatric patients with SCD. While retinal and imaging biomarkers hold potential for assessing systemic disease burden, they are not widely documented for use as clinical indicators of systemic severity, specifically in children with SCD.[10]

Elevated TCD velocities, indicative of increased risk for cerebrovascular events, and decreased Hgb levels are systemic biomarkers that could parallel retinal alterations documented through advanced noninvasive imaging techniques, such as OCT and OCT angiography (OCTA). The current study aims to assess the association of retinal imaging and systemic findings in children with SCD.

Methods

Study Population

Using the same population of a previous study published by our group,[11] this retrospective consecutive cohort study included 303 children with SCD (International Classification of Diseases 10 codes: H36 and H35 and International Classification of Diseases 9 codes: 282.60-282.69 and D57.0-D57.8) aged ≤18 years who had an ophthalmic examination at Boston Children's Hospital between January 1998 and August 2022. The study was approved by the institutional review board at Boston Children's Hospital (protocol ID #P00047894), and the requirement for informed consent was waived because of its retrospective nature. This study adhered to the tenets of the Declaration of Helsinki.

Data collection encompassed demographics, sickle genotype (HbSS, HbSC, HbS thalassemia, and sickle trait), and both retinal and systemic findings. Retinal examination data included best-corrected visual acuity (BCVA) in logarithm of the minimum angle of resolution, the presence and severity of peripheral sickle cell retinopathy, and OCT and OCTA (Spectralis, Heidelberg OCT2, Heidelberg Engineering) imaging findings. Retinal imaging findings comprised total retinal thickness on macular OCT, VD in the superficial capillary plexus (SCP) and DCP, and superficial foveal avascular zone area measured on 6 × 6-mm OCTA scans. All OCT and OCTA parameters were obtained from the automated displays of the imaging devices, and the accuracy of the measurements was reviewed by S.H. to ensure reliability. Systemic data included the number of hospitalizations in the prior year, Hgb and HgbF levels, and TAMV in the right middle cerebral artery (RMCA) and left middle cerebral artery (LMCA) TCD. Time-averaged mean velocity was categorized as low risk (<160 cm/second), conditional (161–185 cm/second), or abnormal (≥186 cm/second). Hydroxyurea treatment and strokes were recorded as binary variables. Specifics regarding qualitative and quantitative analyses of retinal images were described prior in detail.[11]

Statistical Analysis

Outcomes were studied and analyzed using mean and standard deviation for continuous variables and percentages for categorical variables. Spearman correlations were carried out to measure the relation between systemic and ocular variables. Linear mixed models were used to confirm these correlations, as well as to compare retinal imaging parameters and eye examination findings between different genotypes, adjusting for age and race (fixed effects) and eye laterality (random effect). Another analysis was conducted to assess differences by genotype. The statistical significance level was set at $P < 0.05$. SPSS version 28 (SPSS Inc) was used to perform all analysis and descriptive statistics.

Results

Demographic Data and Baseline Characteristics

A total of 606 eyes from 303 pediatric patients (53% males) with SCD were included in this study. Most patients were of Black race (72.3%) and of not Hispanic ethnicity (71.9%). The mean age at presentation was 8.84 ± 5.00 years (median, 9.76; range, 0.1–18). The distribution by SCD genotype was as follows: HbSS (46.2%), sickle trait (25.4%), HbSC (17.5%), and HbS thalassemia (8.3%). Before presentation to the ophthalmic clinic, patients had a medical evaluation with their hematologist or pediatrician. Seventy-six patients (25.1%) had ≥1 hospitalization within 12 months before their ophthalmic visit, with a mean number of hospitalizations of 1.24 ± 0.52 in this subgroup. Regarding other systemic data, 214 patients (70.6%) had available Hgb level (mean: 9.63 ± 1.59 g/dL), and 147 patients (48.5%) had available HgbF level (mean: 0.13 ± 0.11 g/dL). Transcranial Doppler was performed on 137 patients (45.2%), with quantitative results of TAMV in the RMCA and in the LMCA available for 134 patients (44.2%). The mean TAMV in the RMCA and in the LMCA were 115.4 ± 32.7 cm/second and 118 ± 31.0 cm/second, respectively. Time-averaged mean velocity in the RMCA and in the LMCA was >160 cm/s in 11 patients (8.2%) (mean 195.6 ± 27.3 cm/s), and in 13 patients (9.7%) (mean 183.0 ± 28.0 cm/s), respectively. Time-averaged mean velocity in the RMCA and in the LMCA was abnormal (>185 cm/s) in 7 patients (5.2%) (mean 205.7 ± 29.6 cm/s) and 4 patients (3.0%) (mean 217.1 ± 27.2 cm/s), respectively. Regarding eye examination, the mean BCVA at presentation was 0.12 ± 0.23. OCT angiography images were excluded from the analysis if motion artifacts were present that precluded reliable image analysis. In total, OCT and OCTA images were selected for 104 (17.2%) and 60 (9.9%) eyes at presentation and for 159 (26.2%) and 100 (16.5%) eyes at the final visit, respectively.

Correlations between Ophthalmic and Systemic Findings

At initial presentation, BCVA showed a positive correlation with mean Hgb levels ($\rho = 0.098$, $P = 0.048$). After adjusting for age and race, this positive association remained statistically significant ($\rho = 1.264$, $P = 0.006$). However, there was no association between BCVA at presentation and mean HgbF levels, TCD velocities, number of strokes, or number of hospitalizations.

Regarding OCT findings, retinal thinning was associated with a greater number of hospitalizations when measured at the fovea, superior parafovea, nasal parafovea, inferior parafovea, temporal parafovea, superior perifovea, nasal perifovea, inferior perifovea, and temporal perifovea. There was a positive association between retinal thinning and a history of strokes when measured at the fovea and the inferior perifovea. There was also a positive association between retinal thinning treatment with hydroxyurea, when measured at the fovea, superior parafovea, nasal parafovea, inferior parafovea, and temporal parafovea. When studying other markers of systemic severity, retinal thickness on OCT was positively correlated with TAMV in the RMCA at the fovea, inferior parafovea, temporal parafovea, and temporal perifovea. Similarly, retinal thickness was positively correlated with TAMV in the LMCA at the fovea, superior parafovea, inferior parafovea, temporal parafovea, superior perifovea, inferior perifovea, and temporal perifovea. Moreover, foveal retinal thickness on OCT was positively correlated with Hgb level ($\rho = 0.265$, $P = 0.002$) (Table 1). Overall, increased arterial flow velocities below pathologic threshold and elevated Hgb levels are associated with greater retinal thickness across various retinal regions.

Similarly, OCTA findings were correlated with systemic findings. Lower VD of the SCP in the superotemporal and inferotemporal macula were associated with a higher number of hospitalizations. Lower VD of the DCP in the inferotemporal macula was associated with a higher number of strokes. There was no association between VD and hydroxyurea treatment. Regarding TCD velocity, a higher VD of the DCP in the inferotemporal macula positively correlated with TAMV in the RMCA and in the LMCA. Similarly, VD of the DCP in the inferonasal macula positively correlated with TAMV in the RMCA and in the LMCA. Furthermore, lower VD was related to higher Hgb levels. In the superonasal macula, VD of the SCP and DCP was lower in patients with high Hgb levels (Table 2). Overall, these vascular findings were consistent with those related to retinal thickness.

When adjusting for age and race, OCT retinal thinning in the nasal parafovea ($\beta = 0.018$, $P = 0.037$), superior perifovea ($\beta = 0.053$, $P = 0.005$), and inferior perifovea ($\beta = 0.028$, $P = 0.039$) remained positively correlated with the number of hospitalizations. Similarly, OCT retinal thickness in the temporal parafovea ($\beta = 1.870$, $P = 0.011$) and temporal perifovea ($\beta = 0.912$, $P = 0.031$) remained significantly positively correlated with TAMV in the RMCA.

Correlations between Ophthalmic and Systemic Findings by Genotypes

The relationship between eye examination and imaging findings was analyzed according to the specific SCD genotype. Specifically, higher Hgb levels correlated with higher prevalence ($\rho = 0.237$, $P = 0.037$) and severity ($\rho = 0.299$, $P = 0.008$) of retinopathy in HbSC, while it correlated with lower prevalence ($\rho = -0.183$, $P = 0.004$) and severity ($\rho = -0.185$, $P = 0.004$) of retinopathy in HbSS genotypes. In addition, BCVA did not correlate with TCD velocity, Hgb level, or number of hospitalizations in HbSS or HbSC genotypes.

When evaluating OCT findings between genotypes, differences were noted between HbSS and HbS-Thal genotypes. In children with the HbSS genotype, retinal thickness was positively correlated with TCD TAMV in the LMCA, with correlation coefficients of $\rho = 0.272$ at the fovea ($P = 0.018$), $\rho = 0.252$ in the inferior parafovea ($P = 0.028$), $\rho = 0.276$ in the temporal parafovea ($P = 0.016$), and $\rho = 0.264$ in the temporal perifovea ($P = 0.021$). However, for children with the HbS-Thal genotype, there was a negative correlation between TCD TAMV in the LMCA and retinal thickness at the fovea ($\rho = -0.837$, $P = 0.088$), superior parafovea ($\rho = -0.97$, $P = 0.001$), nasal parafovea ($\rho = -0.956$, $P = 0.003$), inferior parafovea ($\rho = -0.91$, $P = 0.012$), superior perifovea ($\rho = -0.97$, $P = 0.001$), inferior perifovea ($\rho = -0.956$, $P = 0.003$), and temporal perifovea ($\rho = -0.956$, $P = 0.003$).

Regarding OCTA findings by genotype, larger foveal avascular zone in the SCP correlated negatively with TCD TAMV in the RMCA ($\rho = -0.365$, $P = 0.04$) and in the LMCA ($\rho = -0.370$, $P = 0.044$) for patients with HbSS genotype. This significant correlation was not observed with other genotypes (HbSC and HbS-Thal).

Discussion

This study constitutes the first dataset of children with SCD correlating ocular findings with systemic outcomes. Our major findings include (1) retinal thinning is associated with a greater number of hospitalization, strokes, and hydroxyurea treatment, (2) in the temporal macula, lower VD was associated with a higher number of hospitalizations and a higher number of strokes, and (3) higher Hgb levels correlated with higher prevalence and severity of retinopathy in HbSC, while it correlated with lower prevalence and severity of retinopathy in HbSS genotypes.

Despite advances in diagnosis and management, SCD still leads to considerable morbidity and mortality even in the pediatric population.[12] Scientific literature on OCT and OCTA findings in children with SCD and their use in clinical practice remains scarce.[2,13–15] Yet the optical biopsy of the retinal tissue offers a unique window into the human body, allowing for the noninvasive study of the retinal microvascular environment.[16] Furthermore, there is a

Table 1. Correlation between Retinal Thickness on OCT Imaging and Systemic Outcomes

OCT Parameter	Number of Hospitalizations	Number of Strokes	Treatment with Hydroxyurea	Time-Averaged Mean Velocity in Right Middle Cerebral Artery	Time-Averaged Mean Velocity in Left Middle Cerebral Artery	Mean Hemoglobin Level	Mean Hemoglobin F Level
OCT total retinal thickness							
Central 1-mm circle							
Correlation coefficient	−0.19	−0.218	−0.188	0.258	0.306	0.265	0.091
P-value	0.016*	0.011	0.023	0.018*	0.005*	0.002*	0.377
Superior parafovea							
Correlation coefficient	−0.28	0.074	−0.159	0.077	0.222	0.048	0.134
P-value	0.000*	0.220	0.047	0.485	0.042*	0.586	0.191
Nasal parafovea							
Correlation coefficient	−0.238	−0.118	−0.206	0.167	0.207	0.158	0.048
P-value	0.003*	0.108	0.015	0.129	0.058	0.072	0.639
Inferior parafovea							
Correlation coefficient	−0.352	−0.135	−0.225	0.291	0.301	0.086	0.025
P-value	0.000*	0.079	0.009	0.007*	0.005*	0.331	0.811
Temporal parafovea							
Correlation coefficient	−0.359	−0.101	−0.227	0.273	0.328	0.156	0.163
P-value	0.000*	0.146	0.008	0.012*	0.002*	0.077	0.114
Superior perifovea							
Correlation coefficient	**−0.212**	−0.058	−0.013	0.181	0.257	−0.074	−0.040
P-value	0.007*	0.271	0.447	0.100	0.018*	0.406	0.697
Nasal perifovea							
Correlation coefficient	−0.173	−0.134	0.041	0.153	0.128	−0.054	−0.084
P-value	0.029*	0.081	0.336	0.165	0.246	0.541	0.414
Inferior perifovea							
Correlation coefficient	−0.166	−0.165	0.028	0.201	0.239	−0.067	−0.125
P-value	0.036*	0.042	0.386	0.067	0.028*	0.447	0.226
Temporal perifovea							
Correlation coefficient	−0.286	−0.102	−0.118	0.261	0.318	0.047	0.068
P-value	0.000*	0.143	0.107	0.016*	0.003*	0.597	0.509

*Statistically significant.

Table 2. Correlation between Vascular Density from OCTA Imaging and Systemic Outcomes

OCTA Parameter	Time-Averaged Mean Velocity in Right Middle Cerebral Artery	Time-Averaged Mean Velocity in Left Middle Cerebral Artery	Mean Hgb	Mean HgbF	Number of Hospitalizations	Number of Strokes
OCTA						
VD in SCP - Total						
Correlation coefficient	0.260	0.35	−0.293	−0.189	−0.188	−0.031
P-value	0.096	0.027*	0.023*	0.208	0.044*	0.741
VD in SCP - ST						
Correlation coefficient	0.199	0.263	−0.166	−0.126	−0.205	0.068
P-value	0.206	0.101	0.204	0.404	0.028*	0.465
VD in SCP - SN						
Correlation coefficient	0.036	0.199	**−0.335**	−0.166	0.017	0.009
P-value	0.820	0.219	0.009*	0.270	0.859	0.925
VD in SCP - IT						
Correlation coefficient	0.212	0.241	−0.156	−0.212	−0.203	0.039
P-value	0.171	0.129	0.231	0.152	0.030*	0.678
VD in SCP - IN						
Correlation coefficient	0.206	0.132	−0.248	−0.170	−0.164	−0.057
P-value	0.185	0.411	0.054	0.255	0.079	0.540
VD in DCP - Total						
Correlation coefficient	0.096	0.325	−0.197	−0.111	−0.029	0.056
P-value	0.547	0.041*	0.132	0.464	0.758	0.550
VD in DCP - ST						
Correlation coefficient	−0.080	0.170	−0.008	0.100	−0.181	0.130
P-value	0.614	0.294	0.951	0.508	0.053	0.159
VD in DCP - SN						
Correlation coefficient	−0.244	0.042	−0.33	−0.006	0.051	−0.060
P-value	0.119	0.795	0.010*	0.969	0.587	0.517
VD in DCP - IT						
Correlation coefficient	0.328	0.342	0.022	−0.116	−0.116	−0.189
P-value	0.032*	0.029*	0.869	0.436	0.215	0.040*
VD in DCP - IN						
Correlation coefficient	0.359	0.374	−0.140	−0.225	0.023	−0.021
P-value	0.018*	0.016*	0.280	0.129	0.805	0.819
Superficial FAZ						
Correlation coefficient	−0.269	−0.233	−0.079	0.238	0.026	0.154
P-value	0.085	0.147	0.549	0.111	0.783	0.657

DCP = deep capillary plexus; FAZ = foveal avascular zone; Hgb = hemoglobin; HgbF = fetal hemoglobin; IN = inferonasal; IT = inferotemporal; OCTA = OCT angiography; SCP = superficial capillary plexus; SN = superonasal; ST = superotemporal; VD = vessel density.
The reported correlations are between OCTA images at the initial presentation and systemic variables, including time-averaged mean velocity in the right and left middle cerebral arteries, mean Hgb levels, mean HgbF levels, and the number of strokes. However, the correlations for the number of hospitalizations are based on OCT findings at the final visit.
*Statistically significant.

significant gap in the available data that explores the correlation between ocular and systemic findings. Understanding this relationship is important, as it could provide insights into how systemic conditions may influence ocular health and vice versa, leading to improved diagnostic and treatment strategies.

Pertaining to the correlations with systemic findings, the study revealed that OCT metrics were associated with an increased risk of hospitalization, strokes, and treatment with hydroxyurea.

Retinal thinning appears to indicate a more advanced stage of SCD. Similar patterns were found when examining VD on OCTA as a retinal marker. In contrast, nonimaging examination findings, including visual acuity, showed only a negligible correlation with the Hgb level ($\rho = 0.098$) and no correlation with mean TCD velocities or the number of hospitalizations. In the literature, 1 study reported that all SCD patients with silent cerebral infarcts exhibited arteriovenous tortuosity on fundus examination.[17] However, the study did not explore further comparisons with other systemic complications or different retinal imaging modalities. Another study used OCTA imaging to analyze the vasculature in 34 SCD patients with silent cerebral infarcts and found reduced VD in the temporal hemifield of the DCP, which supports the current findings.[10] Our work was the first to study a variety of systemic complications in patients with SCD and their relation to retinal vascular biomarkers.

In terms of TAMV, thickness at the fovea, inferior parafovea, temporal parafovea, and temporal perifovea was

positively moderately correlated with the TAMV in both the RMCA and LMCA, as well as with Hgb levels. Although the literature indicates that a high velocity above 200 cm/second is associated with an increased stroke risk, most of the patients in our population had velocities below this threshold, which may explain this finding.[7]

Literature has shown that sickle cell genotypes are linked to different patterns of retinal manifestations. Each genotype may influence the type and severity of retinal involvement, with some genotypes being more prone to central maculopathy and others showing a higher incidence of peripheral retinal changes, highlighting the need for genotype-specific management in affected individuals.[18,19] Results from this current study revealed that the association between eye examination and imaging findings also differed based on the specific SCD genotype. Higher Hgb levels were correlated with a greater prevalence and severity of retinopathy in HbSC genotypes, whereas they were associated with a lower prevalence and severity of retinopathy in the HbSS genotypes. This supports the hypothesis that peripheral sickle cell retinopathy may be associated with higher blood viscosity in HbSC while paramacular atrophy may be related to the severity of the systemic disease, resulting in a chronic low perfusion of the temporal paramacular watershed zone in HbSS genotypes, consistent with previous literature.[20–23] When examining OCT metrics, retinal thickness exhibited a positive correlation with TCD TAMV in patients with the HbSS genotype, whereas these correlations were negative in those with the HbS-Thal genotype.

The integration of quantitative measures from retinal imaging, particularly from OCT and OCTA, as systemic biomarkers in SCD has the potential to make a clinical impact. First, by establishing correlations between ocular findings and systemic disease severity, clinicians may be able to identify patients at higher risk for adverse systemic events based on their retinal vascular morphology,[16,24] which could prompt earlier treatment to prevent complications, including stroke. Similarly, using OCT and OCTA metrics as biomarkers can influence the timing of referrals to ophthalmology and hematology, allowing for earlier intervention when necessary. This is particularly important for pediatric patients, as they are in a critical developmental stage where timely treatment of systemic issues can prevent long-term complications. Understanding the relationship between systemic manifestations and retinal findings has implications for functional vision, as early detection and management of ocular issues could enhance visual outcomes. Additionally, these metrics may enable providers to consider and monitor the efficacy of emerging therapeutic interventions, including gene therapy,[25] both in practice and clinical trials with objective data. The correlation between retinal and systemic health underscores the potential of enhancing personalized care in this population. In the context of emerging gene therapy and systemic therapy, these metrics may guide treatment decisions by identifying patients who would benefit most and the potential implications on visual health.[26]

The current study has several limitations, primarily resulting from its retrospective design. Additionally, there are no age-matched and race-matched controls assessed using OCT and OCTA. To address this, we employed linear regression and mixed models, adjusting for patients' age and race. Furthermore, not all 606 eyes had both longitudinal OCT and OCTA images, and they did not have complete and comprehensive systemic findings such as TCD measurements and Hgb levels in their records. There was also not enough data to monitor patients and their response to treatment. Additionally, we did not record the total number of patients who underwent OCT and OCTA imaging or the proportion of scans excluded due to poor quality, limiting our ability to assess the overall imaging success rate in this cohort. Finally, all OCT and OCTA parameters were obtained from the automated outputs of the imaging devices, which may not provide the same level of accuracy as manual measurements performed with external software. However, the accuracy of the measurements was reviewed by S.H. to ensure reliability.

Thinning of the retina, particularly in the temporal macula, on OCT and loss of VD on OCTA, particularly in the DCP, were correlated with the severity of systemic disease in children with SCD. Notably, compared with visual acuity, retinal imaging parameters were found to have a stronger correlation with the number of hospitalizations and strokes. This suggests that quantitative measures obtained from multimodal retinal imaging could provide valuable noninvasive biomarkers for assessing systemic disease activity and risk in pediatric patients with SCD, enabling early detection and monitoring of systemic complications.

Footnotes and Disclosures

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Abbreviations and Acronyms:
BCVA = best-corrected visual acuity; **Hgb** = hemoglobin; **HgbF** = fetal hemoglobin; **DCP** = deep capillary plexus; **LMCA** = left middle cerebral artery; **OCTA** = OCT angiography; **RMCA** = right middle cerebral artery; **SCD** = sickle cell disease; **SCP** = superficial capillary plexus; **TAMV** = time-averaged mean velocity; **TCD** = transcranial Doppler; **VD** = vessel density.

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